BRAC UNIVERSITY

Inspiring Excellence

Course Title	CSE320: Data Communications
Assignment No.	3
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Section	27
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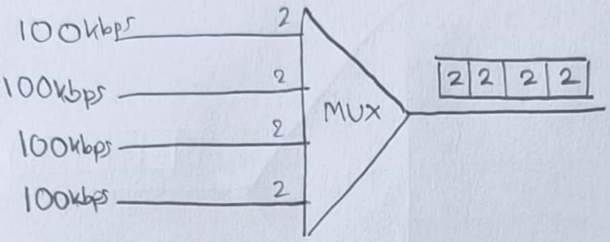
Answer To The Question No. - 1



$$\text{Frame duration} = \frac{2}{100\times1000} \cancel{\text{fps}}$$

$$= \frac{1}{50000} \cancel{\text{fps}} = 20\times10^{-6}\,\text{s}$$

$$\text{Frame rate} = \left(\frac{1}{50000}\right)^{-1} \text{fps} = 50000 \text{ fps}$$

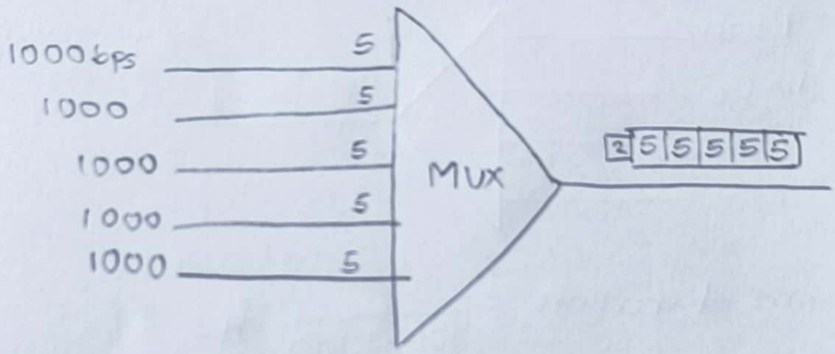
$$\text{Bit}\cancel{\text{rate}}\,\text{duration} = \left(\frac{1}{100\times1000\times8}\right)\text{s} \;\cancel{= 1\times10^{-5}\,\text{s}}$$

$$= 2.5\times10^{-6}\,\text{s}$$

$$\therefore \text{Bitrate} = (50000\times8)\,\text{bps}$$

$$= 400\,\text{kbps}$$

Answer To The Question No. - 2



(a)

Input rate $= 1000$ bps

$= 1$ kbps

Input duration $= \left(\frac{1}{1000}\right)$ s

$= 1 \times 10^{-3}$ s

(b) Frame size $= (5 \times 5) + 2$

$= 27$

Frame ~~rate~~ duration $= \left(\frac{5}{1000}\right)$ s

$= 5 \times 10^{-3}$ s

Frame rate $= (5 \times 10^{-3})^{-1}$ fps

$= 200$ fps

(c) Output rate $=$ (Frame size $\times 200$) bps

$= 5400$ bps

Output duration $= (5400)^{-1}$ s

$= 1.85 \times 10^{-4}$ s

Ans. To The Question No. - 3

Input rate $= 1000$ pages/ 5 seconds

$= (1000 \times 20 \times 40 \times 8)$ bits/ 5s

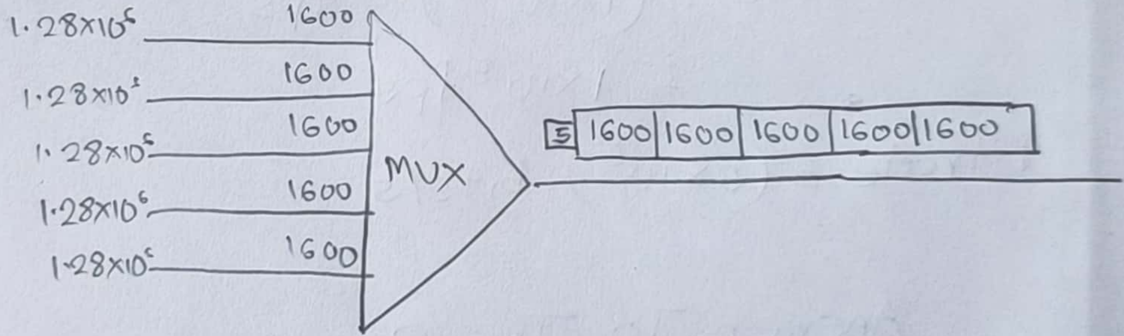
$= 1.28 \times 10^{6}$ bits/s

$= 1.28 \times 10^{6}$ Mbps

Interleaved ~~beat~~ bits $=$ 5 lines

$= (5 \times 40 \times 8)$ bits

$= 1600$ bits



(a) Input rate $= 1.28 \times 10^{6}$ bps $= 1.28$ Mbps

Input duration $= (1.28 \times 10^{6})^{-1}$ s $= 7.8125 \times 10^{-7}$ s

(b) Frame size $= \{(1600 \times 5) + 5\}$ bits $= 8005$ bits

Frame duration $= \left(\dfrac{1600}{1.28 \times 10^{6}}\right)$ s $= 1.25 \times 10^{-3}$ s

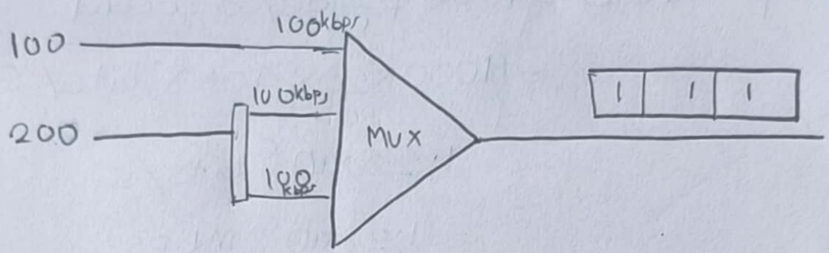
Frame rate $= (1.25 \times 10^{-3})^{-1}$ fps $= 800$ fps

(c) Output rate $= (800 \times 8005)$ bits/s

$= 6.404$ Mbps

Output duration $= (6.404 \times 10^{6})^{-1}$ s $= 1.56 \times 10^{-7}$ s

Answer To The Question No. - 4



Frame duration $= \left(\frac{1}{100 \times 1000}\right) s = 1 \times 10^{-5} s$

Frame rate $= (1 \times 10^{-5})^{-1}$ fps

$= 1 \times 10^{5}$ fps

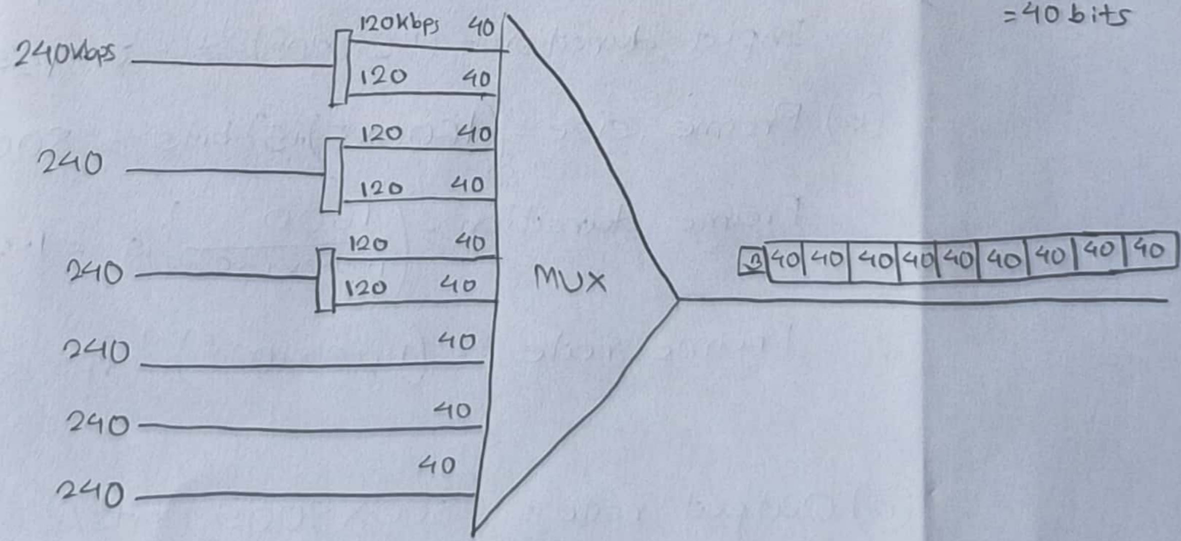
Bitrate $= (3 \times 10^{5})$ bits/s

Answer To The Question No. - 5

(a)

5 characters $= (5 \times 8)$

$= 40$ bits



(b) Input rate = 120 kbps

Input duration $= \left(\frac{1}{120 \times 1000}\right) s = 8.33 \times 10^{-6} s$

(c) Frame size $= \{(40 \times 9) + 3\}$ bits

$= 363$ bits

Frame duration $= \left(\frac{40}{120 \times 10^3}\right)$ s

$= \left(\frac{1}{3000}\right)$ s

Frame rate $= 3000$ fps

(d) Output rate $= (3000 \times 363)$ bits/s

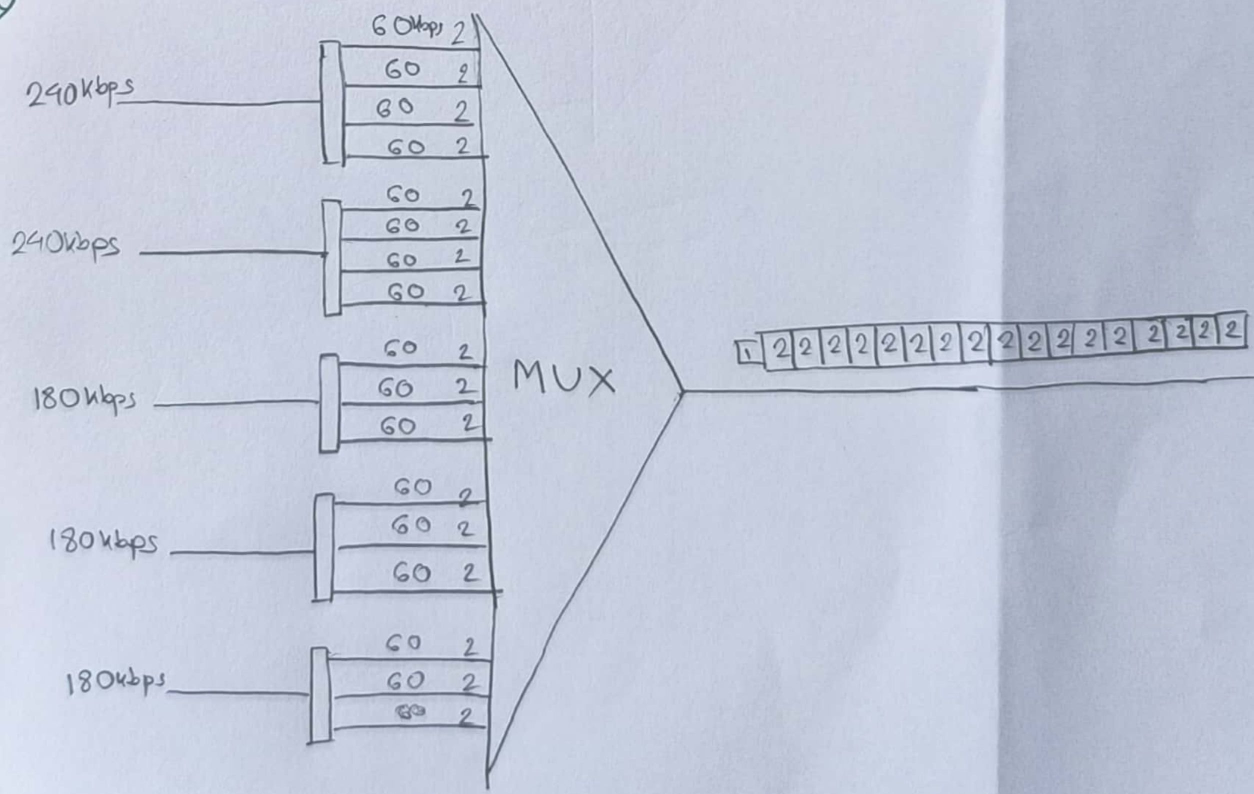
$= 1.089 \times 10^6$ bits/s

Output duration $= (1.089 \times 10^6)^{-1}$ s

$= 9.183 \times 10^{-7}$ s

Ans. To The Question No. - 6

a)



b) Input rate $= 60$ kbps

Input duration $= \left(\frac{1}{60 \times 1000}\right) s = 1.67 \times 10^{-5} s$

c) Frame size $= (17 \times 2 + 1)$ bits

$= 35$ bits

Frame duration $= \left(\frac{2}{60 \times 1000}\right) s = \left(\frac{1}{3 \times 10^4}\right) s$

Frame rate $= 3 \times 10^4$ fps

d) Output rate $= (3 \times 10^4 \times 35)$ bits/s

$= 1.05$ Mbps

Output duration $= (1.05 \times 10^6)^{-1} s$

$= 9.524 \times 10^{-7} s$

Ans. To The Question No.-7

(a)

Overhead for D.S0 = 0

Overhead for DS-1 = $\{1.544 \times 10^3 - (64 \times 24)\}$ kbps

$= 8$ kbps

Overhead for DS-2 = $\{6.312 \times 10^3 - (4 \times 64 \times 24)\}$ kbps

$= 168$ kbps

Overhead for DS-3 = $\{44.736 \times 10^3 - (7 \times 4 \times 64 \times 24)\}$ kbps

$= 1368$ kbps

Overhead for DS-4 = $\{274.176 \times 10^3 - (6 \times 7 \times 4 \times 64 \times 24)\}$ kbps

$= 16128$ kbps

(b) Overhead for ~~ds~~ DS-2 = 168 kbps

DS-2 link framerate = 21000 fps

$\therefore$ Framing bits per frame $= \left(\dfrac{168 \times 1000}{21000}\right)$ bits

$= 8$ bits

Ans. To The Question No. -8

k = Dataword length = 2

n = Codeword length = 8

$\therefore C(n,k) = C(8,2)$

$\therefore$ Redundant ~~bea~~ bits = (8-2)

= 6

Possible datawords = 2^k

= 4

Possible codewords = 2^n

= 2^8 = 256

Valid codewords = 4

Invalid codewords = (256-4)

= 252

Hamming distance:

11100000
00011101
10101010
01010111

d(11100000, 00011101) = 7

d(11100000, 10101010) = 3

d(11100000, 01010111) = 6

d(00011101, 10101010) = 6

d(00011101, 01010111) = 3

d(10101010, 01010111) = 7

$\therefore d_{min} = 3$

$\therefore$ Detectable errors = 3 - 1 = 2

$\therefore$ Error correction capability $= \frac{3-1}{2} = 1$

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$i = 8 \% 4 = 0$

$\therefore$ Codeword = (0+1)th = 11100000

erred codeword = 10011100

Hamming distance here is 5. However the scheme can detect upto 2 errors. So, it cannot be detected. Also, as 10011100 is not a valid code and reciever will discard it.

Answer To The Question No-9

$X = 1338$

$\therefore D = X \% 256 = 58 = 111010$

$G = X \% 16 = 10 = 1010$

```
1010 | 111010000 | 1
       1010
       ----
        1001       1
        1010
        ----
          1100     1
          1010
          ----
           1100    1
           1010
           ----
            1100   1
            1010
            ----
             110
```

$\therefore$ Codeword $= 111010110$

Codeword = 111010110

If the 5th MSB is flipped: 111000110

Now,

```
1010 | 111000110 | 11
       1010
       ----
1010 | 1000
       1010
       ----
         1001
         1010
         ----
1010 | 1010 | 1
       1010
       ----
          0
```

As remainder is 0, the CRC scheme can't detect this error

Ans. To The Question - 10

Found from (9),

$$D = \underset{5}{1}\underset{4}{1}\underset{3}{1}\underset{2}{0}\underset{1}{1}\underset{0}{0} = x^5 + x^4 + x^3 + x$$

$$G = \underset{3}{1}\underset{2}{0}\underset{1}{1}\underset{0}{0} = x^3 + x$$

$$x^3(x^5 + x^4 + x^3 + x) = x^8 + x^7 + x^6 + x^4$$

$$x^3 + x \,\big)\, x^8 + x^7 + x^6 + x^4 \,\big(\, x^5$$
$$x^8 + x^6$$
$$\overline{\quad\quad\quad\quad}$$
$$x^3 + x \,\big)\, x^7 + x^4 \,\big(\, x^4$$
$$x^7 + x^5$$
$$\overline{\quad\quad\quad\quad}$$
$$x^3 + x \,\big)\, x^5 + x^4 \,\big(\, x^2$$
$$x^5 + x^3$$
$$\overline{\quad\quad\quad\quad}$$
$$x^3 + x \,\big)\, x^4 + x^3 \,\big(\, x$$
$$x^4 + x^2$$
$$\overline{\quad\quad\quad\quad}$$
$$x^3 + x \,\big)\, x^3 + x^2 \,\big(\, 1$$
$$x^3 + x$$
$$\overline{\quad\quad\quad\quad}$$
$$x^2 + x$$

$\therefore$ Codeword $= x^8 + x^7 + x^6 + x^4 + x^2 + x$

If the 5th bit from LSB is flipped:

The codeword will be $x^8+x^7+x^6+x^2+x$

Now,

$$x^3+x \,\Big|\, x^8+x^7+x^6+x^2+x \,\Big|\, x^5$$
$$x^8+x^6$$
$$x^3+x \,\Big|\, x^7+x^2 \,\Big|\, x^4$$
$$x^7+x^5$$
$$x^3+x \,\Big|\, x^5+x^2 \,\Big|\, x^2$$
$$x^5+x^3$$
$$x^3+x \,\Big|\, x^3+x^2 \,\Big|\, 1$$
$$x^3+x$$
$$x^3+x \,\Big|\, x^2+x+x$$

∴ As the remainder is not 0, the CRC scheme can detect this error.

Ans. To The Q. No -11

Error detection: ~~We~~ when retransferring the data is comperactively easy and error ~~data~~ rate is low.

Error correction: when retransferring the data is difficult and costly ~~and~~ so errors are corrected on the pathway.

Ans. To The Q. No. -12

This protocol refers to CSMA (Carrier sense Multiple Access)

Maximum propagation time = $(t_4 - t_1) = 1.8\mu s$

$\therefore$ Vulnerable time = $1.8\mu s$

$\cdot$ Frame Transmission time = $t_5 - t_1 = 4\mu s$

For pure Aloha vulnerable Period = $2 \times 4\mu s = 8\mu s$

For Sorted Aloha vulnerable period = $4\mu s$

If the channel bandwith is 10 Mbps,

Minimum Transmission time = $(2 \times 1.8)\mu s = 3.6\mu s$

Minimum frame size = $3.6 \times 10^{-6} \times 10 \times 10^{6} = 36$ bits

A

Ans. To The Q. No - 13

0 - 4 = idle

4 - 10 = Transmission

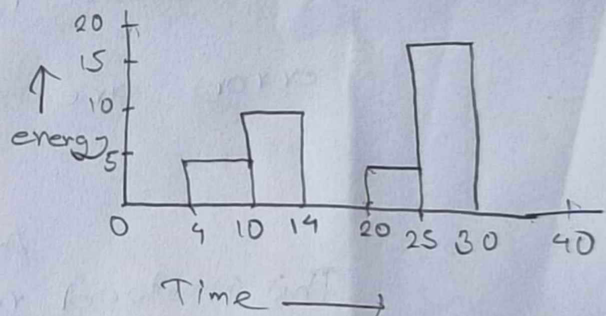
10 - 14 = tramsmission

14 - 20 = idle

20 - 25 = transmission

25 - 30 = collision

30 - 40 = idle



Now,

Transmission = (10 - 4) + (25 - 20) + (14 - 10)

= 15s

idle = (4 - 0) + (20 - 24) + (24 - 30)

= 20s

Collisions = (30 - 25)s = 5s

∴ Available = (0, 4), (14 - 20), (30 - 40)